TASK 5: EXPLORATORY DATA ANALYSIS (EDA)

Titanic Dataset

Objective

To perform Exploratory Data Analysis on the Titanic dataset and identify important patterns, trends, relationships, and anomalies.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

1. Dataset Description

The Titanic dataset contains information about passengers who travelled on the Titanic.

The analysis focuses on variables such as:

Passenger class

-Sex

-Age

-Fare

Survival status

The objective is to understand the factors associated with passenger survival.

2. Data Understanding

The dataset was examined using shape, columns, info(), describe(), and

value_counts() to understand its structure and characteristics.

4. Missing Value Analysis Missing values were identified in some columns of the Titanic dataset.Numerical missing values such as age were handled using the median,while categorical missing values such as embarked were handled using the mode. The deck column was removed because it contains a large number of missing values and is not required for the main analysis.

5. Univariate Analysis

- Survival Count

- Passenger Class
- Age Histogram
- Age Boxplot

5. Univariate Analysis

Univariate analysis involves analyzing one variable at a time. In this analysis, survival status, passenger class, and age distribution are examined using countplots, histograms and boxplots.

5.1 The graph shows the number of passengers who survived and did not survive. The number of passengers who did not survive is higher than the number who survived.

5.2 The graph shows the distribution of passengers among the three passenger classes. Third-class passengers form a large proportion of the dataset.

5.3 The histogram shows the distribution of passenger ages. Most passengers are concentrated in the young and middle-age groups, while fewer passengers are present at very high ages.

5.4 The boxplot shows the median, spread, and possible outliers in passenger age. It helps identify passengers whose ages are unusually high or low compared with the rest of the dataset.

6. Bivariate Analysis
    - Survival vs Gender
    - Survival vs Passenger Class
    - Age vs Fare Scatterplot

6. Bivariate Analysis

Bivariate analysis is used to examine the relationship between two variables. In this analysis, survival is compared with gender and passenger class. The relationship between age and fare is also examined using a scatterplot.

Observation

The graph shows a relationship between gender and survival. Female passengers had a higher survival rate compared with male passengers.

Observation

The graph shows that passenger class is related to survival. Passengers in higher classes generally had better survival outcomes, while third-class passengers had a lower survival proportion.

Observation

The scatterplot shows the relationship between passenger age and fare. Fare values vary considerably among passengers of different ages. The survival status also shows variation across the plotted observations.

7. Multivariate Analysis
    - Correlation Heatmap
    - Pairplot

7. Multivariate Analysis

7.1 Correlation Heatmap

Observation

The correlation heatmap shows the strength and direction of relationships between numerical variables.

7.2 Pairplot

Observation

The pairplot shows relationships and distributions among multiple numerical variables and helps identify patterns in the dataset.

Key Findings

Gender is an important factor associated with survival.

Passenger class shows a relationship with survival.

The age distribution contains passengers from different age groups.

Fare values vary considerably among passengers.

Missing values were identified in some columns.

Correlation analysis helped identify relationships between numerical variables.

Visualizations helped identify patterns and trends in the dataset.

Conclusion

Exploratory Data Analysis was successfully performed on the Titanic dataset using Python, Pandas, Matplotlib and Seaborn.

Different statistical methods and visualizations were used to understand the dataset. Histograms, boxplots, scatterplots, pairplots and correlation heatmaps helped identify patterns, relationships and possible anomalies.

The analysis shows that factors such as gender and passenger class are important in understanding survival outcomes.